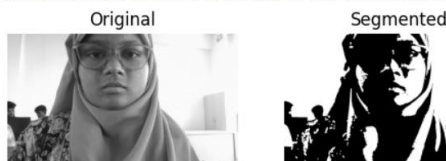


EXP:16

```
import cv2; from google.colab import files; from matplotlib import pyplot as plt
img = cv2.imread(list(files.upload().keys())[0], 0)
_, th = cv2.threshold(img, 127, 255, cv2.THRESH_BINARY)
kernel = cv2.getStructuringElement(cv2.MORPH_RECT, (5,5))
result = cv2.morphologyEx(th, cv2.MORPH_OPEN, kernel)
plt.subplot(121); plt.imshow(img, cmap='gray'); plt.title("Original"); plt.axis('off')
plt.subplot(122); plt.imshow(result, cmap='gray'); plt.title("Segmented"); plt.axis('off')
plt.show()
```

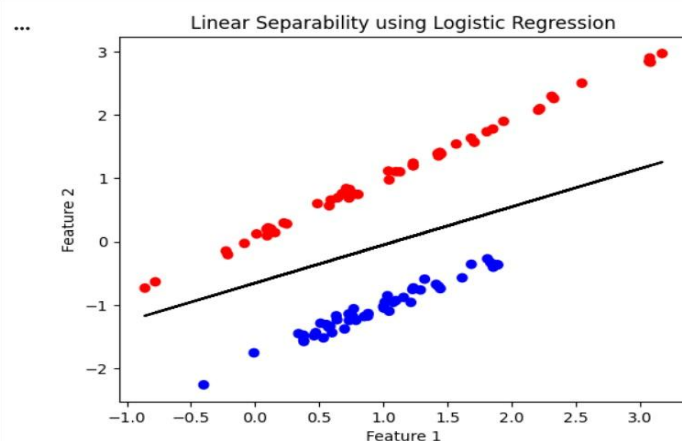
... Choose Files WIN_20260709_10_22_49_Pro.jpg
 WIN_20260709_10_22_49_Pro.jpg (image/jpeg) - 138602 bytes, last modified: 7/9/2026 - 100% done
 Saving WIN_20260709_10_22_49_Pro.jpg to WIN_20260709_10_22_49_Pro (1).jpg



EXP:17

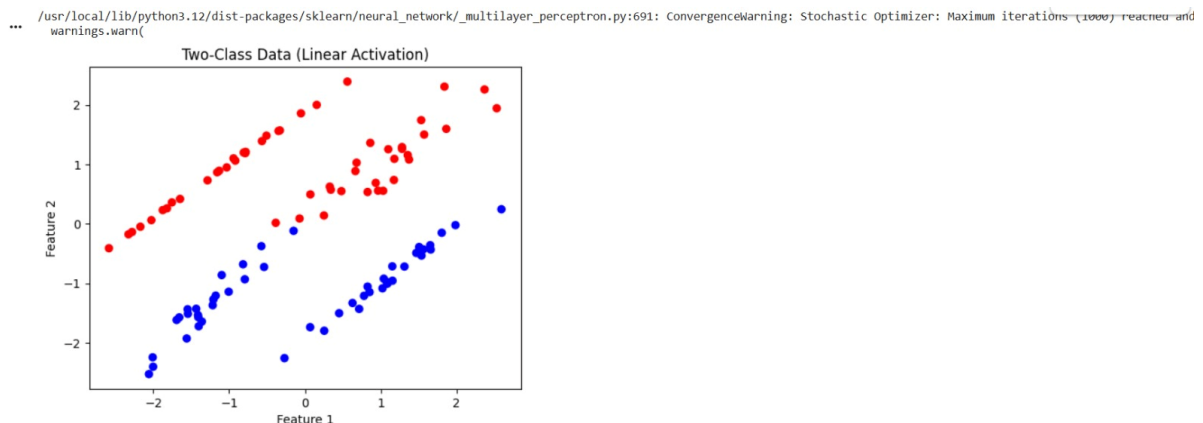
```
from sklearn.datasets import make_classification
from sklearn.linear_model import LogisticRegression
import matplotlib.pyplot as plt

X, y = make_classification(n_samples=100, n_features=2, n_redundant=0,
                          n_clusters_per_class=1, random_state=42)
model = LogisticRegression().fit(X, y)
plt.scatter(X[:,0], X[:,1], c=y, cmap='bwr')
x = X[:,0]
yline = -(model.coef_[0][0] * x + model.intercept_[0]) / model.coef_[0][1]
plt.plot(x, yline, 'k')
plt.title("Linear Separability using Logistic Regression")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.show()
```



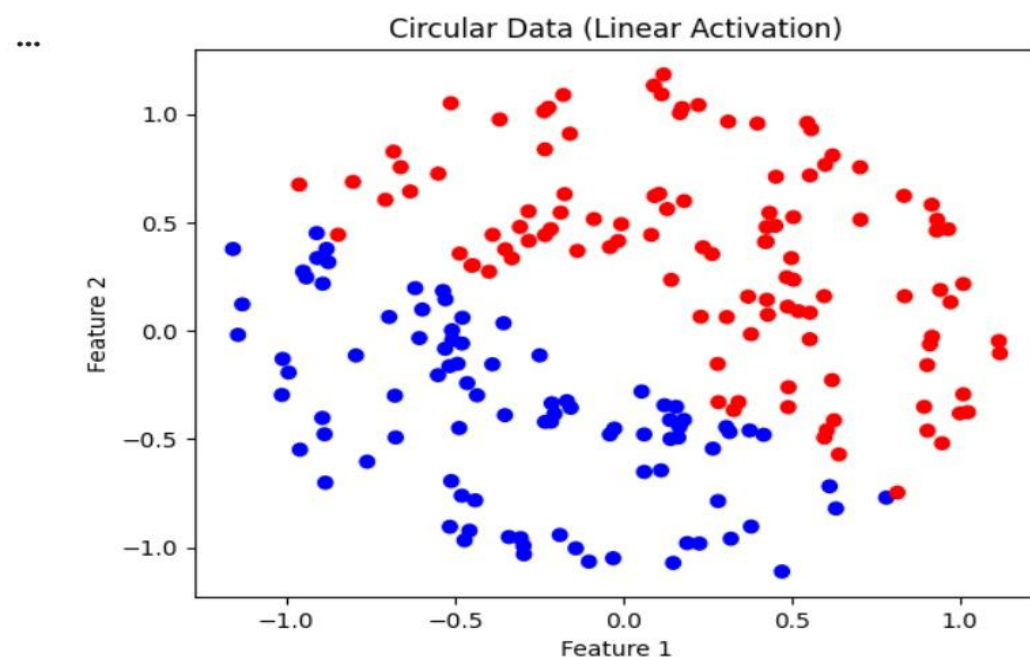
EXP:18

```
from sklearn.datasets import make_classification; from sklearn.neural_network import MLPClassifier
import matplotlib.pyplot as plt
X, y = make_classification(n_samples=100, n_features=2, n_redundant=0, random_state=42)
model = MLPClassifier(hidden_layer_sizes=(), activation='identity', max_iter=1000).fit(X, y)
plt.scatter(X[:,0], X[:,1], c=model.predict(X), cmap='bwr')
plt.title("Two-Class Data (Linear Activation)")
plt.xlabel("Feature 1"); plt.ylabel("Feature 2")
plt.show()
```



EXP:19

```
from sklearn.datasets import make_circles; from sklearn.neural_network import MLPClassifier
import matplotlib.pyplot as plt
X, y = make_circles(n_samples=200, noise=0.1, factor=0.5, random_state=42)
model = MLPClassifier(hidden_layer_sizes=(), activation='identity', max_iter=1000).fit(X, y)
plt.scatter(X[:,0], X[:,1], c=model.predict(X), cmap='bwr')
plt.title("Circular Data (Linear Activation)")
plt.xlabel("Feature 1"); plt.ylabel("Feature 2")
plt.show()
```



EXP:20

```
from sklearn.datasets import make_blobs; from sklearn.neural_network import MLPClassifier
import matplotlib.pyplot as plt
X, y = make_blobs(n_samples=200, centers=3, n_features=2, random_state=42)
model = MLPClassifier(hidden_layer_sizes=(), activation='identity', max_iter=1000).fit(X, y)
plt.scatter(X[:,0], X[:,1], c=model.predict(X), cmap='viridis')
plt.title("Multi-Class Data (Linear Activation)")
plt.xlabel("Feature 1"); plt.ylabel("Feature 2")
plt.show()
```

...

